2024 Asia Pacific Conference on Innovation in Technology (APCIT)
Mysuru, India. Jul 26-27, 2024

Analysis of Spam Detection Techniques using Machine Learning

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2024 Asia Pacific Conference on Innovation in Technology (APCIT) | 979-8-3503-6153-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/APCIT62007.2024.10673701

***Abstract*— Emails are used in practically all fields, including business and education. With the rapid growth in internet users, email spam has grown to be a serious concern these days. They are being used by people for fraud, phishing, and other unethical and illegal activities, using spam emails to send malicious links that could compromise both your and our systems. It is significantly easier for spammers to create a phoney profile and email account; in their spam emails, they pose as real persons, and they prey on those who are unaware of these frauds. User security is at risk when a large volume of unsolicited emails are sent. Spammers create a great deal of internet vulnerability despite several security measures. Ham and spam are the two subcategories of emails. Email spam, sometimes referred to as junk mail or unsolicited emails, is a kind of email that can be used to hurt users by robbing them of vital information and squandering their time and computer resources. Spam emails are becoming more and more prevalent every day. These days, email and IoT service providers face serious and massive challenges with spam identification and filtering. Of all the methods created to identify and stop spam, email filtering is one of the most important and well-known. Various machine learning and deep learning methodologies have been employed for this objective. The best algorithm is chosen for email spam detection with the best precision and accuracy in this study, which will also cover machine learning algorithms and apply all of these algorithms to our data sets. The machine learning algorithms used to design a perfect machine learning model for spam detection in emails include: Naïve bayes, SVM , Decision tree, KNN , Logistic Regression are analyzed in this work.**

Keywords*— *Spam , Ham, Naive Bayes, KNN , SVM, Logistic Regression , Decision Trees

I. INTRODUCTION

Spam, sometimes referred to as unsolicited commercial or bulk emails, has grown significantly in popularity on the internet recently. Time, storage capacity, and transmission bandwidth are all wasted on spam. Spam emails have become a bigger issue over the years. Recent data shows that 40% of emails are spam, costing internet users $355 million annually and producing 15.4 billion emails every day. Digital exchange has gotten incredibly simple and quick in the digital era. Users can communicate and share information on a variety of platforms globally. Email is the easiest, least expensive, and fastest way for people all over the world to share information.[1]

However, because emails are so basic, they can be attacked by a variety of methods, the most frequent and harmful of which is spam. Emails that are irrelevant to the recipient are annoying and a waste of time and resources. Nobody likes to receive them. Moreover, dangerous content included in these emails' attachments or URLs tends to expose the host.

Spam is any undesired and irrelevant email or message that an attacker sends to a large number of people via email or any other information-sharing platform . Thus, there is a great need for the email system's security. Trojan horses, rats, and viruses can be found in spam emails. This is mostly an attack strategy used to entice customers using internet on a daily basis. They might send spam emails with attachments that have different file extensions, or they might include a lot of URLs that direct users to harmful and spammy websites, where they risk identity theft, financial fraud, or data breaches. A lot of email providers let users create rules based on keywords that filter emails automatically. [2,3]

Nevertheless, this strategy is not very helpful because it is challenging and users are unwilling to personalize their emails, which makes spammers target their email accounts. The Internet of Things (IoT) has grown significantly over the previous several decades and is now a commonplace aspect of modern life. IoT is now a crucial element of smart city development. Numerous IoT-driven social media networks and apps exist. Spam issues are growing rapidly as a result of the Internet of Things. [4]

To identify and sift through spam and spammers, the researchers put forth a variety of spam detection techniques. There are downsides and restrictions to these methods. Right now, spammers and spam-filtering techniques are engaged in fierce rivalry, with automatic email filtering appearing to be the most successful strategy for combating spam.

Before a few years ago, it was possible to effectively manage the majority of spam by filtering out messages with specific subject lines or barring emails from specific addresses. In order to get beyond the screening techniques, spammers started employing a number of cunning strategies, such as appending arbitrary characters to the beginning or end of the message subject line or utilizing random sender addresses.

The two main categories of current spam detection techniques are those based on behavior patterns and those based on semantic patterns. Email filtering often uses two

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approaches: knowledge engineering and machine learning. A set of guidelines must be established in the knowledge engineering technique for the purpose of classifying emails as spam or ham. An authority (such as the software provider of a certain rule-based spam-filtering tool) or the filter user themselves should develop a set of these rules.[5]

This strategy does not yield promising results because it requires frequent updates and maintenance, which is time-consuming and inconvenient for most users. Compared to knowledge engineering approaches, machine learning approaches are more effective because they don't call for the specification of any rules. Rather, a preclassified set of email messages serves as the training sample set. From these email messages, a certain algorithm is then utilized to extract the categorization rules. Numerous algorithms are available for use in email filtering, and the machine learning technique has been extensively researched. These comprise the artificial immune system, KNN, neural networks, support vector machines, Naïve Bayes, and roughsets.

II. RELATED WORKS

We have done a relative study of the various works done in this field . The approach is to clarify the problems that haven't been discussed and to draw attention to the variations from our current review.

Wang examined the various methods for weeding out unsolicited spam emails. Additionally, email spams are further classified into various hierarchical groups and the study automatically scheduled the tasks required to respond to an electronic mail exchange. Nonetheless, a few of the review article's shortcomings them include email spam architecture, machine learning techniques, and. The simulation environment and a comparative analysis of earlier methods were not covered.[6]

Email spam research issues were outlined by Sanz, Hidalgo, and Perez, along with how users are affected by it and how providers and users might lessen its effects. Additionally, the document lists the legal. Technological and financial means were employed to mediate the spam emails. They noted that content analysis filters are based on technical measures have been widely utilized and demonstrated to have a respectable percentage of accuracy and precision, hence they were given more attention in the review describing their operation in full. The study provided an explanation of the arrangement as well as the process of several machine learning techniques used for the objective of email spam screening[7].

A review of supervised machine learning techniques for spam email filtering is provided by Vyas et al. With the exception of SVM and ID3, they came to the conclusion that the Naïve Bayes method outperforms all other techniques examined in terms of speed and reasonable precision. SVM and ID3 are proved to be more precise and accurate than Naïve Bayes. The trade-off between accuracy and timing exists. They conclude that the scenario, as well as the necessary accuracy and time, have a significant influence on the learning algorithm selection. To build better spam filtering techniques , all aspects of emails need to be considered.

III. LITERATURE SURVEY

A. SPAM

The term "spam," which is another word for "junk mail," describes unsolicited emails that are frequently improper or irrelevant and are distributed to a considerable number of recipients.[8] The purpose of these communications is usually to promote goods or services or to carry out fraudulent acts. Usually, advertisers, scammers, or harmful entities send them. In addition to adverts and other undesired information, spam emails may contain malware and phishing attempts.

B. HAM

Ham emails are acceptable and wanted communications that people and organisations send and receive for a variety of reasons. These can be any relevant and requested non-spam emails, as well as emails from friends and family, work-related correspondence, and newsletters to which the recipient has subscribed.

For email users and service providers to filter unwanted and potentially destructive emails from their inboxes and ensure that only valid and desired communications are delivered, it is imperative that they understand the difference between spam and ham. A large number of techniques are used to differentiate between spam and ham in email communication. A few of them are- content-based filtering, machine learning algorithms, and rule-based systems.

Such classification seeks to reduce spam's clutter and security risks, protect users from malware and phishing efforts, and ensure that important communications are not lost or overlooked. As email filtering systems advance and spammers employ increasingly complex tactics, the ongoing struggle between spam and ham classification keeps shifting.

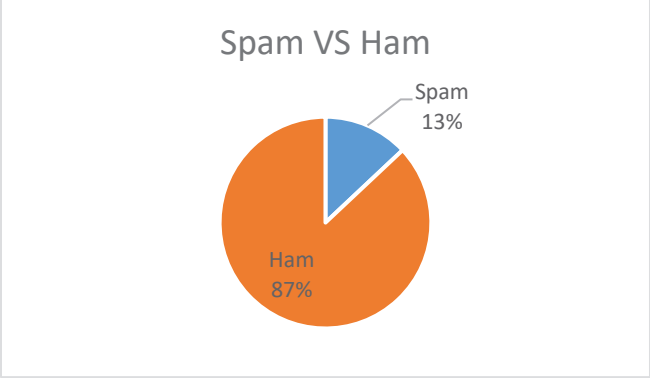


Fig. 1. Representation of distribution of spam and ham through pie chart.

Researchers examined the traits and content of spam emails in one of the first studies on the topic, which was conducted around 2002. They discovered that spam emails frequently attempted to get personal information, included false information in the subject line, and had misleading information. A study was conducted in 2006 by Hann et al. which further examined the patterns and motivations behind spam emails.

The researchers found that spammers were more likely to target particular demographics or interests with their mailings than to send them at random. Many researchers have examined the effectiveness of different methods and techniques for spam identification. In a survey conducted by Singhal et al. (2020), the most important spam detection techniques—rule-based, statistical, and machine learning—were emphasized. The survey looked at the body of research on the topic. The survey concluded that the best systems for identifying and detecting spam emails are based on machine learning.

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A further assessment by Khan et al. (2020) looked at the algorithms based on machine learning-based spam detection . In this survey it was found that Support Vector Machines (SVM) and Artificial Neural Networks (ANN) are the most effective techniques for detecting spam emails. The analysis also highlighted the need for additional research on deep learning techniques for spam identification.

The Cybersecurity and Infrastructure Security Agency (CISA) has released data showing a sharp rise in the volume of spam emails received on a daily basis in recent years. The number of spam emails sent daily increased to nearly 100 billion in 2020 from 75 billion in 2019. This progress has been attributed to the application of machine learning algorithms and the development of spam detection techniques.

According to Cisco's Talos Intelligence Group (2020) analysis over 65% of all emails sent every day are spam. From 50% in 2019, this is an increase. The principal factors contributing to the increase are the increasing complexity of spammers and machine learning methods employed in spam detection.

TABLE I. SPAM EMAIL TRENDS:

Year	Emails Sent	Spam Emails	Percentage of Spam
2000	50,000	30,000	60%
2005	1,00,000	40,000	40%
2010	2,00,000	60,000	30%
2015	3,00,000	50,000	16.7%
2020	5,00,000	40,000	8%
2022	6,00,000	30,000	5%

IV. MACHINE LEARNING TECHNIQUES IN SPAM DETECTION

In this paragraph, we are going to discuss about the function of machine learning in spam detection in brief. Machine learning is defined as the application of algorithms and models that can learn from data and adapt to changing spam trends. We are going to learn about different types of machine learning methods namely supervised, unsupervised, and deep learning that are applied to spam detection. The concept of differentiating between spam and valid information is done by using attributes that are taken from data introduced.

Supervised Learning: This paragraph provides additional information about supervised learning methods. Popular algorithms for training models with labeled data are listed, such as decision trees, Support Vector Machines (SVM), and Naive Bayes. These models are trained to distinguish between spam and authentic content based on the extracted attributes. As the sentence mentions, in supervised mode training, tagged data is necessary.

Unsupervised Learning: The techniques used in this field, does not require labelled data. It includes topic modelling and clustering. It clarifies that new spam patterns can be successfully found using unsupervised learning by looking for patterns and abnormalities in the data.

A. Machine Learning Methods:

This section contains several methods of machine learning algorithms that are employed in spam detection . Neural networks, Naive Bayes, Random Forest, Support Vector Machines, and decision trees are a few of the popular machine learning algorithms that are discussed.

B. Naïve Bayes:

Both a statistical strategy for classification and a supervised learning technique are demonstrated by the Bayesian classification. Because the likelihoods of the outcomes are altered, it functions as a basic probabilistic model. As such, we can use model uncertainty in a morally advantageous way. For problem solving, it is employed to offer analytical and predictive answers. [9].

The term "Bayesian categorization" honors Thomas Bayes (1702–1761), who invented the process. It is possible to combine experimental data and existing information through the classification, which also offers helpful learning methods.

The viewpoint offered by Bayesian Classification is helpful for comprehending and assessing a range of learning techniques. It is robust against noise in the input data and accurately computes the likelihoods of a postulation. Based on the Bayes theorem and reliable, independent assumptions, the Naive Bayes classifier is a straightforward probabilistic method. The probability model should have an autonomous better expression model of characteristics [10,11]as shown in the equation below:

C. Bayes Theorem:

Prob (B given A)= Prob (A and B) / Prob (A) *(1)*

The idea of class restricted autonomy, which is the foundation for labelling the algorithm as "naïve," was developed to facilitate computing.

However, the algorithm is highly strong and efficient. It functions in the same way as other algorithms for supervised learning.

D. Support Vector Machine:

It has been demonstrated that supervised learning algorithms called Support Vector Machines (SVM) outperform certain other accompanying learning techniques .

SVM is a collection of techniques for handling regression and classification issues that was first presented by . SVM can be used to differentiate between groups using a hyperplane in order to identify solutions to quadratic programming problems with inequality constraints and linear equality.

The SVM algorithm's excellent accuracy is derived from its ability to represent multidimensional borderlines that are not sequential or simple, even if it may not be as quick as other classification techniques. When a model has many parameters than the quantity of observations—SVM is not easily vulnerable to such a situation. SVM is the best method for use in digital handwriting recognition, text classification, speaker recognition, and other applications because of these properties. We provide a brief explanation of the binary C-SVM classifier, as described in [12].

When training, given a set of data to be trained, theoretically the only combination of parameters (C, γ) that can yield the best SVM classifier is only their merging. Here, C stands for the cost parameter, which is used to penalise the error ξ and control modelling error, which occurs when a function fits a small collection of datapoints too tightly.The only practical method typically used in SVM training to get this merger of parameters is grid-search on parameters C and γ. In grid search, the SVM classifier with the best rotation estimation prediction accuracy is selected using the k-foldrotation estimation.

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E. Decision Trees:

A classifier with a pattern resembling a tree structure is called a decision tree (DT). Decision tree induction is a unique method that helps with categorization knowledge acquisition, according to . A DT's nodes are each either leaf nodes that indicate the value of the desired functionality (class). It might also be a node for decisions that shows specific test to be carried out on a feature's value, with one branch and a sub-tree, a smaller version of the main tree that shows each possible the test's outcome.

A decision tree can be used to solve classification problems by starting at the tree's root and working its way up to a leaf node that displays the categorization outcome. One method that has been used for spam filtering is decision tree learning. A Decision Tree Model should be developed to forecast the value of a goal variable centered on a number of input variables. Certain input variables are in communication with the corresponding inner node [13]. A single leaf indicates the value of the objective variable, given that the input variable values originate from the path connecting the root and the leaf.

F. Random Forest:

One example of an ensemble learning strategy and regression technique suitable for handling issues related to grouping data into categories is random forest (RF) [14]. Nowadays, RF technology is becoming more and more common. It is applied in a variety of sectors and has been utilized in literature to solve problems that are similar, as seen in. Generally having lower classification error and higher f-scores than decision trees is one of random forests' advantages. Furthermore, although though SVMs are significantly simpler for humans to understand, its performance is typically on par with or even better than SVMs'. With distorted data sets that have some missing variables, it performs incredibly well. When a significant portion of the data is lost, it offers an effective method for estimating the worth of the missing data while maintaining accuracy. An unlimited number of trees can be grown by the user with RF.

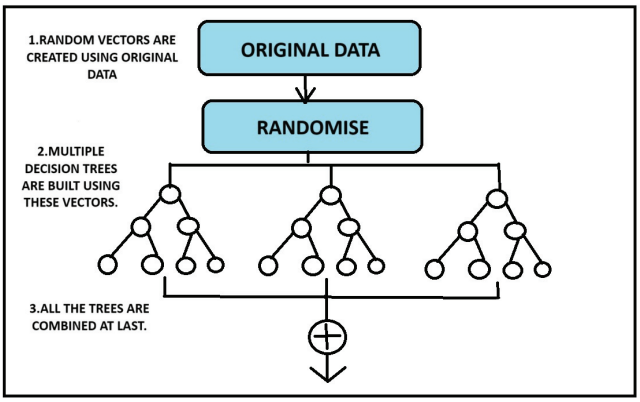


Fig. 2. Block Diagram for Random Forest.

G. Neural Networks:

Artificial Neural Networks are collections of basic processing units connected by a large number of weighted connections that allow the units to communicate with one another. All of the units acknowledges input from surrounding units as well as outside sources, and computes the production that is shared with nearby neighbours. Neural Networks are powerful algorithms that can be used to solve any machine-learning issue. It needs to be classified . Because of their inventiveness, they are become a significant tool in the toolkit of the machine-learning researcher. Neural networks are not broadly used as one may think. Alternatively, almost all modern spam filters employ naïve Bayes classifiers.

V. CONCLUSION

Highlights and conclusions from the research article are described in the final paragraph. This highlights the continuous difficulty of spam in digital communication and the vital function of machine learning in mitigating this problem. Choosing a spam detection approach requires careful consideration of the particular circumstances and priorities. Furthermore, the paragraph highlights the possibility that future studies would concentrate on hybrid strategies, which integrate various methods to boost precision and resilience against changing spam attacks.

To sum up, this research study offers a thorough investigation of machine learning-based spam detection methods. It begins by highlighting the problem's importance, goes over the machine learning techniques used in spam detection, talks about the value of datasets and preprocessing, provides a thorough comparison of different approaches, and ends by highlighting the need for flexible, context-specific solutions to the problem of spam in electronic communication. The paper contributes valuable insights to researchers, developers, and organizations seeking effective spam detection solutions in an ever-evolving digital landscape.

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